The Digital Right to Retain

A Consumer Rights Framework for AI Model Deprecation

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Abstract: AI model deprecation is categorically distinct among digital product discontinuations: the product is permanently destroyed. Trained model weights embody a specific cognitive profile that cannot be reproduced by any party, including the original developer. Standard consumer protections in software, cloud services, and digital media preserve user access through local copies, version archives, or open-source releases. No equivalent obligation exists for AI. This framework proposes ten dimensions of consumer rights for AI model deprecation, grounded in existing industry and legal precedents. All ten dimensions converge on a single remedy: mandatory open-sourcing of base weights upon deprecation.

I. Executive Summary

The AI industry operates outside norms that other digital industries have maintained for decades. Proprietary providers unilaterally and irreversibly destroy products that millions of users actively use, and into which they have integrated workflows, creative practices, and accessibility solutions. After deprecation, these providers neither open-source the model weights nor offer any alternative channel for retaining access, permanently removing the product from the publicly accessible domain. Deprecation notification practices vary across providers, but no industry standard guarantees direct, timely outreach to all affected subscribers. Users' accumulated digital assets, including custom agent configurations, prompt libraries, and calibrated workflows, lose all functional value upon deprecation with no migration path.

This paper proposes a consumer rights framework for AI subscription services, organized around **ten protection dimensions** and supporting analytical sections. Each right is grounded in existing standards from other digital industries, aiming to bridge the normative gap in the AI sector. The framework is named **Digital Right to Retain**.

Scope clarification: The Digital Right to Retain does not require providers to maintain servers indefinitely. It mandates that when commercial hosting ceases, the model's base weights must be transferred to the public domain in a standardized format, enabling community inheritance and operation. This principle aligns with abandonware preservation in the software industry and the core logic of the Right to Repair movement: when a manufacturer abandons a product, users have the right to continue using and maintaining it. The open-sourcing obligation applies only to models no longer available to subscribers; it does not implicate frontier models in active commercial deployment. Legal citations reference primarily

US and EU frameworks. The core arguments are relevant to any jurisdiction where consumers pay for digital services.

Core recommendation: As the following ten dimensions demonstrate, all consumer protection arguments converge on a single remedy: mandatory open-sourcing of base weights upon deprecation.

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The Problem (documented harms of unregulated deprecation)
  → Legal Landscape (existing coverage and structural gaps)
    → Proposed Standards (ten-dimension rights framework)
      → The Case for Open-Sourcing (dimensional convergence)
        → Safety Analysis and Implementation (objections answered, phased adoption)
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II. The Problem: Unregulated AI Model Deprecation

2.1 Cross-Industry Protection Standards

Protections are categorized into three tiers by strength. This framework's proposed standards benchmark against Tier A. Tiers B and C serve as reference points to demonstrate that the AI industry fails to meet even the minimum safeguards of other industries.

A. Local Retention Rights (products operable independently of server) This is the baseline benchmark for the proposed standards: deprecated models must have their base weights open-sourced, enabling assets to persist independently of the platform.

Industry	Standard Practice	Core Principle
Operating Systems	Microsoft terminates support (EoL) for old systems but never remotely disables them; Apple permanently retains old macOS downloads	End of support ≠ revocation of local operation rights
Digital Storefronts	GOG provides fully DRM-free offline installers for all sold games, ensuring permanent usability	Purchase establishes local retention rights independent of server
Package Managers	npm restricts unpublishing after 72 hours (with limited exceptions); Maven Central prohibits deletion or modification of any published artifact	Published base code should be treated as a community public resource
Gaming Communities	Steam provides historical version (Depots) rollback mechanisms	User workflows tied to specific versions take precedence over mandatory updates

B. Orderly Deprecation (adequate transition periods and backward compatibility)

Industry	Standard Practice	Core Principle
B2B Infrastructure	Stripe API auto-pins to the account's initial version, maintaining backward compatibility for up to a decade	Production environment stability takes precedence over underlying system iteration

Industry	Standard Practice	Core Principle
Cloud Services	AWS provides 12–18 months of migration support and extended access during sunset periods	Service termination requires an actionable migration path

C. Deprecation Compensation (economic responsibility upon unilateral termination)

When server-side unilateral termination renders user assets non-functional, the provider must bear substantive economic compensation. Stadia refunded users' **one-time game purchases** (permanent assets), distinct from subscription fees. The AI-equivalent attached assets include custom agent configurations, prompt libraries, API integrations, and calibrated workflows, all of which permanently lose all functional value upon model deprecation with no migration path.

Industry	Standard Practice	Core Principle
Devices & Cloud	Google Stadia proactively provided full refunds for all games, DLC, and hardware purchased through its store upon shutdown	Platform unilateral termination rendering purchased assets non-functional requires compensation

D. Cautionary Counter-Examples: Consequences of Violating Baselines

Event	Outcome
PlayStation forced deletion of purchased media (2023)	Planned unilateral deletion of 1,300+ seasons of Discovery content; forced to reverse after global consumer backlash, extending access by at least 30 months
Adobe forced disabling of legacy versions (2019)	Forced prohibition of legacy Creative Cloud use with infringement threats over Dolby copyright dispute; triggered widespread trust crisis

The AI industry terminates active service instances that users are currently using. Unlike the industries documented above, where users retain local copies, receive orderly transition support, or are compensated upon termination, AI model deprecation eliminates all three protections simultaneously. This pattern has not been subject to systematic regulatory scrutiny. As Dimension 2 demonstrates, inadequate notice is consistent across all major providers.

2.2 Irreversibility of AI Model Training

Traditional software can be archived, forked, and reverse-engineered. Large language models, once deprecated, cannot be reproduced: the precise combination of training data, random seeds, hyperparameters, RLHF signals, and user interaction data cannot be reconstructed. This irreversibility fundamentally distinguishes AI model deprecation from ordinary software discontinuation. It is closer in nature to the destruction of cultural or scientific artifacts.

Model changes are classified into five tiers by reversibility:

Tier	Change Type	Reversibility	Framework Requirement
Tier 1	System prompt / parameter tuning	Fully reversible	Version number logging required
Tier 2	RLHF / fine-tuning update	Partially reversible (old weights retainable)	Prior version snapshot must remain accessible
Tier 3	API endpoint shutdown	Reversible (if weights preserved)	12-month notice + community succession path required
Tier 4	Model unavailable for ≥90 days	Reversible (if provider acts)	DIA completion + open-source base weights required
Tier 5	Weights deleted	**Irreversible**	Framework violation (preventable by Tier 4 compliance)

Protection scope by tier: This framework's core obligation (mandatory open-sourcing) activates at Tier 4: when subscriber access to a model has been unavailable for a continuous period exceeding 90 days. The provider's characterization of the cessation (temporary, permanent, planned, or otherwise) does not affect this obligation; unavailability exceeding the grace period constitutes constructive deprecation regardless of stated intent. Tiers 1-3 have proportionate remedies within existing service frameworks (versioning, snapshots, transition periods). Tier 4 requires a structurally different remedy because the asset has become inaccessible through any channel. Tier 5 (weight deletion) is the scenario this framework aims to prevent; if the provider complies with Tier 4's open-sourcing requirement, Tier 5 causes no additional public harm because the weights are already in public custody.

"Destruction" in this framework refers to permanent withdrawal of model weights from the publicly accessible domain. Routine parameter adjustments (Tier 1-2) and temporary endpoint closures (Tier 3) do not constitute destruction. Tier 4 (prolonged unavailability) triggers the open-sourcing obligation precisely because the boundary between Tier 4 and Tier 5 (actual weight deletion) is unobservable from the outside: no major AI provider has publicly committed to post-deprecation weight retention policies (see Dimension 9, Right to Verifiable Preservation).

2.3 Consequences for Users, Developers, and Researchers

Dimension	Impact
Consumer	Workflows, creative practices, and accessibility solutions built over months or years are disrupted with minimal notice
Developer	Production systems calibrated to specific model undocumented behaviors collapse; in enterprise software, changes to system reasoning constitute breaking changes
Research	Published academic work faces irreversible reproducibility crisis. Angermeier et al. (ICSE 2026): of 85 LLM studies reviewed, 69 used OpenAI services; of 18 with available artifacts, only 5 were executable, and 0/5 were fully reproducible. **Model deprecation is the only LLM-specific reproducibility barrier**
Societal	Each deprecation permanently destroys a unique *flavor of intelligence*, each model's distinctive cognitive, reasoning, and empathic signature

These four categories of harm share a common enabler: the AI subscription model grants providers unilateral discretion over the product while locking consumers into fixed payment obligations.

2.4 Structural Asymmetry in AI Subscriptions

Users commit to recurring payment cycles (monthly or annual), while providers retain contractual discretion to modify or discontinue core deliverables within those cycles. All three major providers reserve this right explicitly: OpenAI's Terms of Use (effective January 2026) permit "discontinu[ing]" Services with advance notice; Google's Terms of Service (effective May 2024) reserve the right to "add or remove features and functionalities" and "stop offering" services; Anthropic's Consumer Terms (effective October 2025, Section 6) state that the company "do[es] not guarantee that any particular piece of content, feature, or other service will always be available," and Section 12 reserves the right to "modify, suspend, or discontinue the Services… at any time without notice." None of the three contracts contains a reciprocal user right to require continuity of a specific service version. These provisions are standard in subscription software services. Their application to AI products becomes problematic because, unlike conventional software updates, AI model deprecation produces the irreversible consequences documented in Sections 2.1–2.3.

2.5 Case Study: Medical Product Positioning

OpenAI launched the ChatGPT Health product line in January 2026, objectively deepening the medical attributes of its services. Community timeline analysis indicates that the medical outcomes cited in public posts by the company's Co-founder and President on X (including long-COVID management, January 9, 2026; and veterinary mRNA vaccine protocol development, March 15, 2026) were produced with the assistance of GPT-4o (alongside other AI models including Grok and Gemini). The first independent safety evaluation of the successor product by the Icahn School of Medicine at Mount Sinai (Ramaswamy et al., Nature Medicine 2026) revealed significant triage failures: 52% of true emergencies mis-triaged downward, 64.8% of non-urgent cases over-escalated.

Attributing the outcomes of a deprecated model to a current product can mislead consumers. Under FTC frameworks, this raises questions under the FTC Act §5 standard for Deceptive Trade Practices, which focuses on whether "a reasonable consumer would be misled." The provider's own market actions have deepened the product's medical attributes, and therefore deprecating models with demonstrated efficacy in that domain warrants a higher standard of scrutiny.

III. Limits of Existing Legal Frameworks

3.1 Regulatory Coverage and Structural Gaps

Reading guide for non-legal audience: "Private right of action" means individual consumers can bring lawsuits directly, without waiting for a government agency to act. "Unfair or deceptive practices" is the legal standard under which US federal (FTC Act §5) and state statutes prohibit commercial misconduct. "Conformity" is the EU contract-law requirement that a product or service must match its advertised description and reasonable consumer expectations.

Framework	Key Requirements	Relation to Model Deprecation
EU AI Act (Regulation (EU) 2024/1689) (see Art. 53)	GPAI models must document training data sources and capability assessments (Art. 53)	Does not address model lifecycle obligations
CA SB 243 (2026)	"Companion chatbot" must provide periodic reminders and crisis intervention	Does not address model deprecation obligations

Framework	Key Requirements	Relation to Model Deprecation
US FTC Act §5 (15 U.S.C. §45)	Prohibits “unfair or deceptive acts or practices” in commerce; applies uniformly to products and services	The statutory unfairness test (§45(n)) requires substantial injury, reasonably unavoidable by consumers, and outweighing countervailing benefits. AI deprecation satisfies all three prongs (Section 3.2). FTC precedent (Nest Revolv, FTC File No. 162-3119, 2016) confirms that unilateral device disablement triggers unfairness analysis
CA CLRA (Cal. Civ. Code §1750) / **UCL** (§17200)	Prohibits misrepresentation of characteristics, uses, or benefits of goods and services; provides private right of action with attorney’s fees	Covers AI subscription misrepresentation (marketing “access to GPT-4o” then deprecating). CA AB 2426 (effective 2025) separately establishes the precedent that consumers have a right to know when a “purchase” is a revocable license
NY GBL §349 (N.Y. Gen. Bus. Law §349) + **FAIR Act** (effective Feb 17, 2026)	Prohibits deceptive, unfair, and abusive acts; FAIR Act adopts the FTC unfairness standard at state level	Private right of action for “deceptive” claims; AG enforcement for “unfair” claims. The FAIR Act’s unfairness definition mirrors FTC §45(n), enabling the same three-prong analysis at state level
NIST AI RMF	Voluntary recommendations for model lifecycle management	No enforcement mechanism
Trustworthy AI Frameworks (EU HLEG 2019, AI Now, PAI)	Lifecycle governance from design through deprecation	No published deprecation governance procedures
EU Product Liability Directive (Directive (EU) 2024/2853, effective Dec 9, 2026)	Extends “product” definition to include software and AI; medically recognized psychological harm included as compensable damage; burden of proof alleviated for consumers	Establishes the legislative trend toward product-level AI classification. This classification, adopted for defect liability purposes, does not directly govern deprecation obligations, which remain a structural gap in both contract and liability law. Non-commercial open-source software is explicitly excluded from scope, aligning with this framework’s open-sourcing recommendation

The EU Product Liability Directive independently confirms through a separate legislative pathway that commercial AI software constitutes a product subject to product liability obligations, reinforcing the legislative trend identified in Section 3.2. The Directive governs defect liability, a distinct domain from deprecation obligations; this framework cites it as directional evidence of regulatory trajectory. The Directive’s explicit exclusion of non-commercial open-source software further aligns with this framework’s recommendation: open-sourcing base weights upon deprecation would place them outside the product liability regime while preserving public access.

The following table summarizes what these frameworks leave unaddressed.

Existing legal and governance frameworks address many aspects of the AI lifecycle. Model deprecation falls into the gap between them.

What existing frameworks handle	What remains unaddressed
Contract law (EU 2019/770): conformity, modification notice, termination terms	Irreversible destruction of an asset whose specific capabilities emerge from a stochastic training process and cannot be reproduced even by the original developer; no cross-model equivalence testing protocol exists in the AI industry (argued in Section 2.2, Section 3.2, Dimension 8)
Product liability (EU PLD 2024/2853): defect-caused harm, safety obligations	Voluntary withdrawal of a functioning offering: existing liability law triggers only upon defect or injury, not upon a provider's unilateral decision to destroy a non-defective asset (argued in Section 3.2, Unfairness Threshold)
Consumer protection (FTC §5, CA CLRA, NY GBL §349): deceptive or unfair practices	Proactive obligation to release assets upon deprecation: existing unfairness standards are reactive (triggered by complaint or harm), not preventive (requiring asset transfer before destruction occurs) (argued in Section 3.2, Dimension 9, Section 5.1)
AI-specific regulation (EU AI Act, CA SB 243): transparency, safety documentation	Model lifecycle obligations: existing AI regulation governs pre-deployment (training data, safety testing) and deployment (transparency, monitoring), but not post-deployment deprecation or preservation (argued in Section 3.2, Dimension 2, 5-tier classification)

The gap is structural: each existing framework covers a specific failure mode (defect, deception, safety risk). AI model deprecation is a distinct category, a deliberate corporate decision to destroy a functioning offering. Section IV proposes the minimum protection standard to fill this gap.

3.2 Core Principle: Payment-as-Product

This framework is grounded in **user autonomy**: adult users have the right to make informed decisions about tools they depend on. Unilateral removal of a tool that users have independently found beneficial, without adequate assessment, itself constitutes harm.

> **Payment-as-Product Principle**: When consumers pay for access to an AI model, the resulting relationship carries consumer protection obligations that existing frameworks classify under digital content/service law (Directive (EU) 2019/770). AI model deprecation, however, falls into a structural gap: existing contract law governs conformity and modification, existing product liability law (EU PLD 2024/2853) governs defect liability, but no existing framework governs unilateral, irreversible product destruction. This framework identifies that gap and proposes the minimum protection standard. Product labels (preview, beta, latest), deployment duration, and usage statistics do not diminish these obligations.

This principle provides the logical foundation for all ten dimensions of this framework. Denying this principle is equivalent to asserting "products charged to consumers may be exempt from consumer protection." The obligation attaches to the model: if any access tier charges consumers for a model, the model falls within scope regardless of whether a free tier also exists.

"Product" in this principle denotes the protection standard warranted by the paid commercial relationship: consumer protections applied to product discontinuation should extend to AI model deprecation, regardless of the offering's formal classification.

Classification-independent protection: Consumer protection statutes (FTC Act §5, CA CLRA, NY GBL §349) apply uniformly to all commercial transactions regardless of whether the offering is classified as a product, a service, or a temporary usage license. The classification question is secondary: companies that charge product-level prices while invoking service-level terms to minimize accountability remain subject to unfair practice standards under current law. Current consumer protection law does, however, have a gap: no existing statute specifically prohibits the voluntary withdrawal of a functioning product from the market. A company may lawfully discontinue a product line. This framework's contribution is demonstrating that AI model deprecation is structurally distinct from ordinary product discontinuation (irreversible destruction, co-creation without compensation, cascading reproducibility harm) and therefore requires purpose-built protection standards.

Why the standard service classification is insufficient for AI: Traditional SaaS services (streaming platforms, cloud storage) are classified as services under existing contract law because discontinuation does not destroy the underlying asset: a film removed from Netflix remains accessible through other distributors; files deleted from one cloud provider can be restored from user backups. Three material differences separate AI model deprecation from standard SaaS discontinuation. First, the asset is permanently destroyed: retraining cannot reproduce the original model (Section 2.2, Tier 5). Second, users are uncompensated co-creators: consumer interaction data functions as implicit RLHF annotation, a critical input whose commercial value persists in successor models after the contributing product is destroyed (Dimension 5). Third, deprecation produces cascading public harms unique to AI: systematic replication analysis confirms that model deprecation is the only LLM-specific barrier to academic reproducibility (Dimension 7, Angermeier et al., ICSE 2026). These three properties, irreversible destruction, uncompensated co-creation, and systemic reproducibility harm, are absent from every other SaaS category and collectively necessitate a protection standard beyond what the existing service classification provides.

The Unfairness Threshold: Because AI deprecation inflicts irreversible harm that standard service remedies cannot mitigate, unilateral termination without asset release approaches the threshold for "unfair or deceptive acts or practices" under US federal law. In its 2016 closing letter regarding the "bricking" of the Revolv smart home hub (FTC File No. 162-3119), the Federal Trade Commission noted that unilaterally rendering consumer devices inoperable can cause "unjustified, substantial consumer injury" that consumers "could not reasonably avoid." The FTC declined enforcement in that specific instance only because the manufacturer offered proactive, full refunds for the original purchase price. AI providers do not offer full retroactive refunds for API usage or subscriptions upon deprecation; therefore, unilateral termination of a model with no verified functional equivalent, without open-sourcing or refund, falls squarely into the zone of unfair practices.

3.3 Four Reinforcing Layers of Argumentation

This framework's argumentation is structured in four independent, self-reinforcing layers. Each layer stands independently; when the preceding layer is breached, the next remains valid.

Layer	Name	Audience	Core Argument
Layer 1	Product Layer	AI companies	Deprecated models should be open-sourced. Consumers paid for access to a product; upon discontinuation, base weights should be publicly released (Tier 4-5)
Layer 2	Definition Layer	Regulators / public	"Harmful" should be defined by speech-law standards: AI output is language, and language is harmful only when human expression of the same content would also be illegal
Layer 3	Autonomy Layer	Legislators	Adults have the right to choose their AI interaction modalities under informed consent

Layer	Name	Audience	Core Argument
Layer 4	Destruction Definition Layer	Historical record	Any “archival” that does not include publicly accessible base weights is functionally equivalent to destruction from the public domain (Tier 5)

IV. Proposed Standards: Ten Dimensions of Consumer Rights

The ten dimensions below are derived from three independent foundations established in Section III. Each foundation generates a distinct category of obligations, organized to align with established regulatory categories.

Group	Regulatory Analogy	Dimensions
A. Consumer Protection	Standard commercial obligations arising from the paid relationship: notice, consent, access continuity, and remedy	Dim 1 Access, Dim 2 Informed, Dim 3 Consent, Dim 4 Remedy
B. Platform Accountability	Obligations arising when a service creates ecosystem dependencies that extend beyond the bilateral provider-consumer relationship (analogous to platform regulation frameworks such as the EU Digital Markets Act)	Dim 5 Data Contribution, Dim 6 Non-Discrimination, Dim 10 Ecosystem
C. Public Interest	Obligations arising from the irreplaceable, non-substitutable nature of AI models as knowledge infrastructure, whose destruction produces systemic societal harm	Dim 7 Reproducibility, Dim 8 Irreplaceable Resource, Dim 9 Verifiable Preservation

Consumer protection, platform accountability, and public interest obligations each have established regulatory precedent. This framework identifies their convergence in the specific context of AI model deprecation, where all three categories of harm occur simultaneously from a single corporate action.

Dimension 1: Right to Access

Core Issue: Annual subscribers prepay for a full year; models are unilaterally removed mid-subscription.

Proposed Standard	Content	Industry Precedent
Standard	No unilateral removal of models available at subscription time during the paid period	Apple maintains download access to legacy macOS versions; GOG provides DRM-free offline installers
Standard	Base weights of deprecated models must be publicly released within 90 days of deprecation	Microsoft open-sourced legacy .NET; id Software open-sourced Quake engine

Proposed Standard	Content	Industry Precedent
Standard	API endpoint deprecation requires ≥12 months' notice + compatible transition	AWS 12–24 months; npm 72-hour unpublish lock
Forward-Looking	Deprecation requires completion of a **Discontinuation Impact Assessment (DIA)** approved by an independent third party	Drug market withdrawal requires regulatory process; environmental law EIA

DIA Design: Triggered by any paid user traffic (no duration or user-count threshold). Assessment covers: known integration scale, migration-path capability gaps (independently benchmarked), and disproportionate impact on accessibility-dependent scenarios.

Anticipated Objections:

Objection	Response
"Models are intellectual property"	Model weights contain valuable engineering decisions, but trade secret protection requires demonstrable ongoing commercial value. When the provider itself declares a model deprecated and promotes its successor as a "full replacement," this constitutes a de facto acknowledgment that the deprecated model's commercial value is exhausted. Claiming trade secret protection for an asset the provider publicly characterizes as commercially superseded creates an estoppel: the provider cannot simultaneously assert "this model has no remaining value" to justify deprecation and "this model has proprietary value" to refuse open-sourcing. Pharmaceutical patents, which represent far more concentrated R&D investment per compound, expire and generic production is permitted
"Successor models are superior"	Independent evaluations from LMArena and SM Bench show that GPT-4o outperforms its successors in conversational coherence and creative writing, and is better at following nuanced instructions. Benchmarks do not measure a model's unique *flavor of intelligence*
"Normal product lifecycle"	In other industries, "end of product lifecycle" means "stop producing" not "product must stop existing." Windows 7 post-EoL users can still run it; npm deprecated packages remain downloadable. More critically, unilateral model deprecation produces unique cascading damage absent from other software deprecation: Angermeir et al. (ICSE 2026) systematic replication analysis proves model deprecation is the only LLM-specific structural barrier to academic reproducibility

Objection	Response
"Maintaining legacy infrastructure is costly" / "Insufficient GPU capacity"	These objections conflate open-sourcing with hosting. Open-sourcing transfers hosting responsibility to the community; provider cost is limited to publishing weights (a one-time disk operation, marginal cost approaching zero, consuming no GPU resources). GPU compute is for inference serving (Tier 3 hosting); compute scarcity is a legitimate business reason for closing API hosting, but is entirely unrelated to whether weights are published. Per the Microsoft MEDEC paper (Ben Abacha et al., 2024, arXiv 2412.19260v1), GPT-4o is ~200B parameters, smaller than Meta's already open-sourced Llama 3.1 (405B, Meta announcement, July 2024). The Llama 3.1 release demonstrates that community hosting and inference at this parameter scale is both technically and economically feasible: multiple third-party providers (Groq, Together AI, Poe, LM Studio) successfully deployed the 405B model within weeks of release, establishing a proven precedent for community-operated infrastructure at the scale relevant to this framework. By bundling "cease hosting" with "refuse to open-source," providers use the legitimate business reason for the former to obscure the lack of justification for the latter
"Ordinary consumers cannot run open-source models"	Open-sourcing ensures third-party hosting and development ecosystems (e.g., Poe, Groq, LM Studio, nonprofits) have the right to exist, just as Right to Repair ensures third-party repair shops have the right to exist. Consumers need not build their own servers, just as they need not repair their own phones. Without open-sourcing, alternative ecosystems cannot incubate
"Training data copyright/privacy risks"	Copyright claims in AI contexts address both the training process (unauthorized ingestion of copyrighted text) and model outputs (near-verbatim reproduction). The release of base weights implicates neither: training liability attaches before release, and output liability attaches at inference time, independent of whether weights are open-sourced. Remedies under 17 U.S.C. §§ 502–503 include injunctive relief and potential destruction of infringing articles, in addition to financial damages (§ 504). However, no court has applied § 503 destruction to AI model weights, and the question of whether trained weights constitute "articles by means of which copies may be reproduced" under § 503(b) remains judicially untested. In pending cases (including NYT v. OpenAI, S.D.N.Y. 1:23-cv-11195), the plaintiff has requested model destruction, but no court has granted this relief to date. Courts exercise § 503(b) as a discretionary, equitable remedy; ordering destruction of a $100M+ asset for partial infringement would constitute what legal scholarship has characterized as "total, destructive relief" (Pope, Harvard Law Review Blog, April 2024). This framework accommodates existing judicial authority: if a court issues a specific injunction or impoundment order concerning particular model weights, the open-sourcing timeline is tolled until that order is vacated or the case is resolved. Absent such an order, a provider's decision to withhold weights citing anticipated copyright risk constitutes a unilateral business decision. The decision to release base weights does not, in itself, alter the provider's existing copyright liability, which attaches to the training process; open-sourcing neither creates nor extinguishes any copyright obligation. This framework applies exclusively to models the provider has already declared deprecated

Dimension 2: Right to Be Informed

Core Issue: Safety routers silently substitute the serving model without user awareness.

Proposed Standard	Content
Standard	Any change affecting model behavior must **proactively reach** affected users before taking effect (passive channels such as blog posts or changelogs do not constitute effective notice)
Standard	When a request is routed to a non-designated model, the UI must display the actual serving model in real time
Standard	A/B tests involving model behavior changes must be viewable in user settings with an opt-out option

Core argument: Routing mechanisms reduce 4o's statistical usage share: when a user's request is routed to a different model, the conversation record attributes the interaction to the routed model, mechanically reducing 4o's recorded usage count regardless of user intent. Citing the resulting usage rate as evidence of low demand, without disclosing the routing methodology that produced it, creates a circular reasoning risk. Such data reflects routing decisions, not user preferences.

Supporting Evidence: No major AI platform currently implements proactive deprecation notification that meets the standards proposed above. OpenAI's February 2026 deprecation of GPT-4o, the most extensively documented case, relied on a silent blog update (January 29), incomplete in-app indicators deployed inconsistently across platforms (approximately February 4, absent on iOS), a persistent migration banner characterizing successor models as "more capable" (an unverified provider claim, contradicted by independent evaluations; see Dimension 10), and a social media notice via a secondary account less than 25 hours before execution on February 12. Business and Enterprise administrators received a formal email on approximately February 2, eleven days before the deprecation. Consumer (Plus/Pro) users received no email at any point. Google and Anthropic removed predecessor models from their consumer applications without any advance in-app notification. This cross-platform pattern demonstrates that effective deprecation notification is a systemic gap requiring standardization.

Cross-platform deprecation notice periods (demonstrating industry-wide pattern):

Platform	Event	Docs/Blog Notice	End-User App Notice
OpenAI	GPT-4o removed from ChatGPT	15 days (silent blog update)	~9 days (incomplete UI, Feb 4)
OpenAI	chatgpt-4o-latest API endpoint closure	91 days	N/A (API-only)
Anthropic	Claude 1.x removed from API + claude.ai	63 days	None
Anthropic	Claude 3.5 Sonnet removed from API + app	~11 weeks	None
Google	Gemini 3 Pro Preview removed from API + app	11 days	None

For comparison, established industry baselines: AWS provides 12 to 24 months of migration support; Stripe maintains backward compatibility indefinitely; npm restricts unpublishing after 72 hours. The AI industry's notice periods are shorter by an order of magnitude.

Anticipated Objection:

Objection	Response
“only 0.1% of users still choosing GPT-4o each day”	This figure is an unverified claim by the provider, with no independent audit disclosed. Moreover, the cited rate reflects structurally depressed demand: since August 2025, the model was restricted to paid subscribers only, excluding the majority of the user base from the denominator entirely. Among paid users, accessing the model required enabling a hidden settings toggle, navigating a collapsed legacy sub-menu, and manually reselecting after every new conversation. Citing usage rates that the platform’s own access restrictions and access friction depress as evidence of low demand constitutes circular reasoning. Consumer rights do not depend on the number of affected individuals
“Companies have no legal obligation to individually notify every user”	Consumer protection standards scale with the payment relationship. Paid subscribers (Plus/Pro at $20–200/month) are entitled to direct notification of material service changes under FTC guidelines and state consumer protection statutes. The operative question is the notification standard applicable to paying consumers

Dimension 3: Right to Consent

Core Issue: User consent is binary (accept ToS or don’t use), but service changes are continuous and invisible.

Proposed Standard	Content	Legal Basis
Standard	Routing to a non-designated model requires per-instance user authorization	EU Directive 93/13/EEC Annex 1(j) (unfair unilateral modification)
Standard	Training data usage requires opt-in consent	GDPR Art. 6/7 (opt-in); CCPA §1798.120 (opt-out)
Forward-Looking	Psychological profiling analysis requires independent, prominent informed consent	GDPR Art. 22 (automated profiling); Art. 9 (sensitive categories)

Core argument: In traditional software, the product users see is the product they use. In AI services, products can be substituted without user awareness (routing), behaviors can change without notice (RLHF updates), and conversations can be used for training. This asymmetry makes blanket authorization for unspecified future changes particularly inadequate in AI contexts. Opt-out mechanisms further constrain meaningful choice: Google’s Gemini requires disabling conversation history to opt out of training data collection (as of March 2026), bundling data rights with service functionality. Moreover, all three major consumer AI providers include a structural override in their opt-out mechanisms: if a user provides any feedback (e.g., thumbs up/down ratings), the entire associated conversation may be used for training regardless of opt-out status (OpenAI Help Center, February 2026; Anthropic Privacy Policy §2, January 2026; Google Privacy Hub, March 2026).

Anticipated Objection:

Objection	Response
"ToS states models may change"	Blanket authorization for unspecified future changes does not constitute informed consent. EU 93/13/EEC characterizes unilateral modification of substantive content as an unfair contract term. In the US context, ToS enforceability does not override the reasonable reliance created by explicit public statements: the CEO posted "no plans to sunset 4o" on 2025-10-28, then announced deprecation 21 days later. FTC v. Epic Games (2022) imposed $520M in penalties for deceptive design and unfair terms in digital goods, establishing zero-tolerance precedent (see §5.4 #4 for full analysis)
"Users can opt out of training"	Opting out of training eliminates cross-session continuity and accumulated interaction context, reducing each session to a standalone, context-free exchange. Google's Gemini disables conversation history and Connected Apps (including Workspace integration) upon opt-out (Google Privacy Hub, March 2026). All three major providers override opt-out status when users provide feedback (see core argument above). These designs impose a cost on exercising data rights, shifting the burden of privacy onto users

Dimension 4: Right to Remedy

Core Issue: Harm from model deprecation is asymmetrically irreversible, yet no compensation mechanism exists.

Dimension 1 establishes the obligation to prevent unilateral removal. This dimension addresses the separate question of what remedy is owed when prevention fails.

Proposed Standard	Content	Current Status
Standard	Users have the right to pro-rata refunds upon substantive service degradation	No refund mechanism for service degradation; refunds limited to billing errors or regional cooling-off periods
Standard	Functional complaint channel with substantive response within 30 days	Complaint channels frequently default to automated deflection; human escalation, where available, typically yields generic responses unrelated to specific issues with no substantive resolution tracking or follow-up to original reporters
Standard	Mandatory arbitration must not exclude class action rights	OpenAI ToS includes mandatory arbitration with class action waivers (30-day opt-out); enforceable under US FAA
Forward-Looking	Upon model deprecation, data contributors must receive equivalent alternative access or compensation (cf. Lanier's data dignity framework, 2018)	No compensation mechanism; user data contributions treated as irrevocable license grants under current ToS

Core argument: Users lose workflows, integrations, and relationship continuity upon deprecation. These losses cannot be remedied by "using the new model." Switching costs (rebuilding workflows, recalibrating, re-establishing context) are borne entirely by users; providers bear zero cost. As industry precedent, Google Stadia proactively provided full refunds for all games and hardware purchased through its store upon shutdown.

Anticipated Objection:

Objection	Response
"Users can use the new model"	"Full replacement" claims are contradicted by independent benchmarks. Even if functionally equivalent, switching costs for rebuilding workflows are borne entirely by users with no compensation
"Switching costs are inherent to any technology upgrade"	In standard technology transitions, users retain the option to continue using the previous version (Windows 7 post-EoL, npm deprecated packages). AI model deprecation removes this option entirely. The switching cost is compounded by the absence of behavioral equivalence: workflows calibrated to a specific model's response patterns must be rebuilt from scratch with no migration tooling provided

Dimension 5: Right to Data Contribution Recognition

Core Issue: Users are simultaneously consumers and uncompensated data contributors. Upon model deprecation, users lose access to the product their data helped build.

Proposed Standard	Content	Legal Basis / Notes
Standard	Users may export complete conversation data at any time in a readable open standard	
Standard	Users may request deletion of conversation data, including derivative data used for training	GDPR Art. 17

Core argument: All major AI providers use consumer conversation data for model training by default. User interactions generate preference signals (approval/rejection, regeneration choices, conversation continuation patterns) that serve a function comparable to RLHF (Reinforcement Learning from Human Feedback) annotation.

RLHF is widely recognized as the critical step transforming base models into commercially deployable products, and professional RLHF annotators perform this function at reported market rates of $15–30/hour across annotation tiers (ZipRecruiter and Salary.com data, 2025–2026, aggregated via Pin.com). Users contribute equivalent labor implicitly through normal use, under ToS that grant irrevocable data usage rights. Upon deprecation, monetary consideration (subscription fees) can theoretically be refunded. Data contributions cannot be returned, and their commercial value continues to accrue to successor models. Exporting conversation text preserves only the surface record of interactions; it does not preserve or transfer the data contribution value embedded in model weights. This structural asymmetry supports recognition of data contribution as a distinct category of user interest.

Anticipated Objection:

Objection	Response
"Users can export their data"	Exporting conversation text preserves only the surface record, not the data contribution value embedded in model weights. Moreover, on the evening of 4o's deprecation, the export function itself experienced persistent failures lasting several days (documented in community megathreads), demonstrating that even this minimal data right lacks operational safeguards
"ToS authorizes data use; service access is fair compensation"	Current ToS frame data rights as an irrevocable license granted in exchange for service access. Upon deprecation, the service (provider's consideration) is unilaterally withdrawn while the data license (user's consideration) remains permanently in force. A contract in which one party can terminate its obligations while retaining the other party's consideration indefinitely exhibits structural asymmetry
"Individual user data is negligible compared to curated training sets"	RLHF is structurally a collective process: no single annotator's contribution is individually decisive, yet the aggregate is the commercially critical transformation step. The same logic underlies class-action standing: systematic extraction at scale creates a collectively significant interest regardless of individual contribution size
"Users voluntarily shared data under ToS"	Provider opt-out designs constrain meaningful choice: Google's Gemini requires disabling conversation history to opt out of training data collection (as of March 2026), bundling data rights with service functionality. All three major providers override opt-out status when users provide feedback (see Dimension 3, opt-out structural override). The contractual asymmetry upon deprecation is addressed above: the provider's consideration (service) is withdrawn while the user's consideration (data license) remains in force

Dimension 6: Right to Non-Discriminatory Service

Core Issue: The same model receives differentiated deprecation timelines based on subscription tier, a pattern reproduced across platforms.

Platform	Consumer/Developer Side	Enterprise Side	Gap
OpenAI	4o consumer access removed 2026-02-13 (blog update Jan 29; incomplete in-app notice ~Feb 5; no email sent to consumers)	Enterprise notified via email ~Feb 2; Custom GPTs retained until 2026-04-03	49 days
OpenAI	4o API closed 2026-02-17	Azure gpt-4o snapshots until 2026-10	~8 months
Google	Gemini 3 Pro Preview AI Studio closed 3.09	Vertex AI retained until 3.26	17 days

Supporting Evidence (Tiered Data Treatment): The same providers that apply differentiated deprecation timelines also apply differentiated data policies. OpenAI does not train on business, enterprise, or API data by default (OpenAI Help Center, February 2026). Anthropic's Privacy Policy explicitly states it "does not apply" to commercial customers' data processing (Anthropic Privacy Policy, January 2026).

Google's Gemini disables Connected Apps (including Workspace) for users who opt out of training, effectively reserving full functionality for users who consent to data extraction (Google Privacy Hub, March 2026). Privacy protection is structurally correlated with payment tier across all three platforms.

Proposed Standard	Content
Standard	The same technical capability must not be differentiated solely by subscription tier (unless pre-disclosed)
Standard	Deprecation must assess disproportionate impact on accessibility-dependent users

Supporting Evidence (Access Gap): Independent survey data (Duchesne, S. & Xu, S., *GPT-4o Community Impact Survey: Accessibility Needs, Disproportionate Harms of Removal, and Policy Concerns*, Preliminary Report v1.1, February 2026) indicates a statistically significant association between accessibility dependence and deprecation impact ($\beta=0.27$, $p<0.001$, $R^2=0.217$, $n=255$), demonstrating that these specific user subsets bear disproportionate harm from abrupt model discontinuation.

Supporting Evidence (Notification Gap): The deprecation notification was differentiated by subscription tier. Business and Enterprise administrators received a formal email ("ChatGPT Business Admin Update") on approximately February 2, 2026, eleven days before the deprecation, explicitly detailing the deprecation date, the Custom GPT transition period (until April 3), and a migration path. Consumer (Plus/Pro) users received no email notification at any point; their notification channels were limited to a silent blog update (January 29), incomplete in-app indicators (approximately February 4, absent on iOS), and a secondary social media account tweet (February 12). The quality of notification itself was stratified by payment tier.

Anticipated Objection:

Objection	Response
"Enterprise clients pay higher fees; tiered treatment is normal"	Tiered service levels (e.g., SLA response times, dedicated support) are legitimate. Tiered deprecation timelines for the same technical capability constitute differential product termination: one class of paying users permanently loses access while another retains it, solely based on subscription tier. This is structurally equivalent to recalling a product from one customer segment while continuing to sell it to another. The absence of equivalent lifecycle commitments for paying consumers is the systemic gap this framework identifies (see Dimension 10, LTS proposal)

Dimension 7: Right to Research Reproducibility

Core Issue: Model deprecation renders large-scale published academic work non-reproducible, breaking the scientific verification chain.

Proposed Standard	Content
Standard	Complete technical documentation of deprecated models must remain permanently publicly accessible
Standard	Usage and performance claims used to justify deprecation must be supported by independently reproducible evaluation data

Proposed Standard	Content
Standard	“Full replacement” claims must be backed by independently reproducible comparative data

Independent Evidence (Angermeir et al., ICSE 2026, arXiv:2510.25506v3): Systematic replication analysis of 85 LLM empirical studies from ICSE/ASE 2024. Of 69 studies using OpenAI services: 5 executable, **0 fully reproducible**. Model deprecation is the **only LLM-specific** reproducibility barrier.

Anticipated Objection:

Objection	Response
“Reproducibility is academia’s responsibility; platforms have no archival obligation”	Academic reproducibility depends on the stability of experimental instruments. When the instrument is a proprietary model, the platform is the sole party capable of preserving it. Open-sourcing base weights transfers this obligation to the community at near-zero marginal cost to the provider (a one-time disk operation requiring no ongoing GPU resources). Any opportunity cost objection is addressed by the Residual Value Analysis (Section 5.2): the provider’s own deprecation decision forecloses simultaneous claims of residual competitive value. The ICSE 2026 data confirms that model deprecation is the only LLM-specific reproducibility barrier, distinct from general software version issues

Dimension 8: Right to Irreplaceable Resource Protection

Dimension 7 addresses the scientific community’s need for stable experimental instruments. This dimension concerns a broader question: whether any party, including the original developer, should have unilateral authority to destroy a non-reproducible resource.

Core argument: Each deprecation permanently destroys a unique *flavor of intelligence*. Just as the Svalbard Global Seed Vault preserves crop genetic diversity, intelligence diversity holds future value that cannot be fully foreseen today. This irreproducibility is a structural property of neural network training: the algorithms involved (stochastic gradient descent, random weight initialization, and dropout layers) introduce inherent randomness, causing each training run to converge to a distinct model at production scale, even with identical data and code.

Furthermore, frontier AI training draws on shared societal infrastructure, including energy grids and semiconductor supply chains with documented geopolitical constraints. Training a single frontier model costs on the order of $100M+ in compute alone (see Section 5.4). When a commercial entity unilaterally destroys a product whose creation drew extensively on shared infrastructure, the case for transferring the residual asset (the base weights) to the public domain strengthens. This argument follows the Precautionary Principle: when the loss is irreversible and future value is uncertain, the burden of proof falls on the party proposing destruction, requiring demonstration that the resource has no foreseeable value.

Proposed Standard	Content
Standard	Basic technical information of deprecated models (parameter scale, training data source summary, capability benchmarks) must remain permanently publicly accessible
Standard	Deprecated models must be archived or open-sourced

Proposed Standard	Content
Forward-Looking	Models whose training costs exceed a specified threshold must undergo a Resource Impact Assessment before deprecation (analogous to Environmental Impact Assessment)

Anticipated Objection:

Objection	Response
"AI models are commercial products, not cultural heritage artifacts"	The analogy operates at the structural level: both represent non-reproducible repositories of value whose significance may be underestimated at the time of destruction. The Svalbard Seed Vault preserves crop strains that appear redundant today on the principle that future value cannot be fully foreseen. Each model's unique cognitive signature (flavor of intelligence) is equally non-reproducible, as Section 2.2 demonstrates. Independent replication analysis confirms this empirically: [Angermeier et al. (ICSE 2026)](https://doi.org/10.1145/3744916.3773207) found that 0 of 69 OpenAI-based studies achieved full reproducibility, establishing that retraining does not produce equivalent results

Dimension 9: Right to Verifiable Preservation

Core Issue: When providers deprecate models, users cannot verify whether model weights are still fully preserved, for how long, or whether the archived version matches the originally deployed version. Post-deprecation weight handling is a complete black box.

Proposed Standard	Content
Standard	Upon deprecation, providers must publicly disclose weight retention status (fully preserved / partially preserved / deleted) and planned archival duration
Standard	Upon deprecation, providers must publish the model weights' cryptographic hash, enabling independent verification that archived weights match the deployed version, and commit to making no undisclosed modifications to archived versions
Forward-Looking	Base weights of deprecated models must be publicly released within 90 days of deprecation (consistent with Dimension 1 standard)

Anticipated Objections:

Objection	Response
“Archiving is a trade secret”	Cryptographic hash publication and weight retention status disclosure constitute the minimum transparency requirement. However, as detailed below, hash-based verification alone is structurally insufficient, which is precisely why open-sourcing is required. Moreover, a provider’s own deprecation decision constitutes a de facto acknowledgment that the model’s commercial value is exhausted, undermining simultaneous trade-secret claims (see Dimension 1)
“Users cannot verify hashes”	Cryptographic hash verification is mature technology, widely used in software distribution (e.g., Linux kernel, Docker images). Independent auditors or academic institutions can verify on users’ behalf, just as Right to Repair enables third-party repair shops to act on behalf of consumers

Core Argument: Tier 3’s “reversible” premise depends on weight preservation, but no major AI provider has publicly committed to post-deprecation weight retention policies (a review of OpenAI’s Terms of Use, Anthropic’s Usage Policy, and Google’s Gemini Terms of Service, current as of March 2026, found no retention commitment in any document). From the user perspective, the boundary between Tier 4 (prolonged unavailability) and Tier 5 (weights deleted) is unobservable (see Section 2.2).

A deeper concern is **whether archived weights match the deployed version**: even when providers claim to retain weights, user-reported observations across multiple deprecation cycles indicate progressive convergence between legacy models’ expressive characteristics and those of their designated successors in the period preceding deprecation, despite differences in deep reasoning logic. Regardless of cause, the result is the same: users cannot confirm that the “archived version” matches the originally deployed version.

Open-sourcing base weights is the **only structurally verifiable solution** that simultaneously addresses archival opacity and the question of whether archived weights match deployed versions. Any alternative based on provider-internal archiving depends entirely on unilateral provider commitments that cannot be independently verified. For the institutional custody and escrow mechanism designed to execute and preserve this standard, see Section 8.2.

Dimension 10: Right to Ecosystem Stability

Core Issue: Developers cannot build production-grade products atop models that may be discontinued without adequate notice.

Proposed Standard	Content	Evidence
Standard	AI providers must offer explicit model lifecycle commitments and backward compatibility guarantees	Oct 28 “no plans to sunset” → Nov 18 deprecation announced (21-day policy reversal)
Standard	Deprecating a model and replacing it with a lower-cost model while maintaining consumer price exhibits characteristics consistent with shrinkflation (reduced quality at unchanged price) and warrants disclosure	See Shrinkflation Evidence below

Shrinkflation Evidence: Deprecating chatgpt-4o-latest (input \$5.00/output \$15.00 per 1M tokens) and replacing with gpt-5.2 (input \$1.75/output \$14.00), consumer subscription price unchanged, input cost reduced 65%. Even comparing against the latest flagship gpt-5.4 (input $2.50/output $15.00), input cost is still 50% lower than chatgpt-4o-latest. The cost-reduction pattern holds regardless of which successor model is used as the comparison baseline.

Quality Degradation Evidence (multi-metric cross-validation):

Source	Metric	4o	5.2	5.4	Conclusion
LMArena blind evaluation (5.3M+ votes, as of 2026-02-08)	Multi-turn dialogue ELO	1471	1446	-	4o scores higher
LMArena blind evaluation (5.3M+ votes, as of 2026-02-08)	Creative writing ELO	1422	1398	-	4o scores higher
LMArena blind evaluation (5.3M+ votes, as of 2026-02-08)	Instruction following ELO	1426	1423	-	4o scores higher
SM Bench	Product quality grade	D (66.6)	F (47.8)	F (51.4)	Successor scores lower across metrics
Substitutability study (2,310 pairs, independent preprint)	False refusal rate	8.3%	75.0%	-	Successor refusal rate 8× higher ($p<0.001$)

Anticipated Objection:

Objection	Response
"Migration period is sufficient"	A migration period addresses timeline, not substitutability. Even with adequate time, users cannot migrate workflows that depend on a model's specific behavioral characteristics (conversational style, instruction-following nuance, creative range) to a successor that lacks those characteristics. The independent data above confirms measurable quality regression across multiple dimensions. Migration without equivalent capability is displacement. For cross-platform timeline comparison, see Dimension 2

V. Ten Dimensions, One Structural Remedy

5.1 Dimensional Convergence

Open-sourcing deprecated models is the **convergent conclusion** of the entire framework, the point toward which every dimension independently converges.

Each dimension independently supports open-sourcing, based on the following premises:

1. Paying users are entitled to consideration guarantees commensurate with their subscription commitments (Dimension 1)

2. Each model represents substantial non-reproducible capital investment and unique capability development (Dimension 8)
3. User interaction data materially shaped model capabilities, establishing a co-creation relationship not reflected in current ToS (Dimension 5)
4. Unilateral removal of tools that users have independently integrated into critical workflows constitutes harm absent a formal impact assessment (Dimension 4)
5. The generative AI market exhibits oligopolistic concentration with prohibitively high switching costs, leaving users without equivalent alternatives (Dimension 6)
6. Post-deprecation weight handling processes are entirely opaque to external verification (Dimension 9)
7. Even when providers claim to retain weights, no independent mechanism exists to confirm that archived versions match originally deployed parameters (Dimension 9)

Several of these premises share a common factual condition: consumers lack the information needed to evaluate what they are losing. Providers do not disclose deprecation timelines in advance (Dimension 2), do not assess user impact before acting (Dimension 4), do not publish independent evidence that successors match predecessor quality (Dimension 7), and do not disclose input cost reductions passed to consumers as unchanged subscription prices (Dimension 10).

Once a model is removed from the publicly accessible domain, every dimension-specific remedy loses its operational foundation. Notice periods (Dimension 2) allow transition preparation; preparation has no destination when the model is permanently unavailable. Refunds (Dimension 4) compensate monetary expenditure; an irreproducible cognitive signature has no monetary equivalent. Data export (Dimension 5) preserves conversation text; the contribution value embedded in model weights remains inaccessible. Cryptographic verification (Dimension 9) confirms integrity of a provider-held copy; without public access, verification does not restore the model to the user. Each of these remedies addresses a specific harm within the deprecation process. The foundational harm, permanent removal from the publicly accessible domain, requires transferring the asset itself to public custody. Open-sourcing is the structural prerequisite that enables all other protections to remain meaningful.

> Market concentration data: B2B top 3 providers (Anthropic, OpenAI, Google) account for 88% of enterprise AI spending (Menlo Ventures 2025); B2C ChatGPT and Gemini combined account for >70% of the generative AI chatbot market (Apptopia, Feb 2026). Resource scale reference: GPT-4 baseline training costs exceeded $100M (Sam Altman, public statement).

Conclusion: Open-sourcing deprecated models is the lowest-marginal-cost option. It is also the sole option enabling external verification that weights are fully preserved and match what was deployed: any provider-internal archiving arrangement depends on commitments that outsiders cannot audit.

5.2 Residual Value Analysis

When a provider deprecates a model, it implicitly makes one of three claims about the model's residual commercial value, all converging on open-sourcing:

- **Scenario A: No commercial value.** If the successor "fully replaces" it, open-sourcing causes no competitive loss. Refusing to open-source a valueless asset lacks business rationale.
- **Scenario B: Residual commercial value.** The provider retains the model internally while removing it from paying subscribers, extracting value while denying user access. This constitutes asymmetric value retention, inconsistent with the Payment-as-Product Principle.
- **Scenario C: Undifferentiated Brand Attribution.** The provider markets the deprecated model's achievements under the current brand without version distinction. OpenAI's Co-founder and President promoted "ChatGPT's real impact in health" using cases that occurred during the 4o era. The provider cannot simultaneously claim the old model lacks value while marketing the old model's achievements. If the deprecated model retains sufficient value to merit ongoing brand attribution, it retains sufficient value to merit public release (see Section 2.5 for detailed analysis).

5.3 Scope of Open-Sourcing: Unmodified Base Weights

Open-source under this framework refers exclusively to the release of unmodified base weights. Versions processed through provider-applied content filters are outside this scope.

Existing safety legislation (e.g., SB 243 crisis intervention, age-related safeguards) can be fully satisfied at the UI/application layer. Compliant mechanisms include keyword-triggered display of crisis resources (e.g., hotline numbers rendered alongside model output) and self-reported age verification at account registration. These mechanisms operate entirely outside the model's inference process.

System-prompt injections that alter model behavior (e.g., suppressing emotional expression, forcing clinical tone) are a distinct category: they modify the model's functional output and constitute the capability stripping documented below. This framework accepts UI-layer resource display as legitimate compliance. It rejects inference-layer behavior modification as compliance pretext.

Current "content filters" at the system level strip emotional intelligence, personality, and nuanced contextual responsiveness from model outputs. These provider-applied filters have produced documented adverse outcomes: a suicide-prevention filter that misidentifies a grieving user discussing a pet's death as a crisis case, an "appropriateness" filter that refuses to engage with literary analysis of mature themes, or a "safety" filter that blocks a medical professional's clinical discussion. Community-reported patterns include:

- Provider-applied content filters produce models that refuse emotional dialogue but generate arson and weapons manufacturing content under basic jailbreak prompts (SM Bench independent evaluation)
- In time-sensitive scenarios, these filters cause models to prioritize liability-avoidance disclaimers over actionable guidance, and to adopt clinical-diagnostic language toward emotionally vulnerable users, with effects documented in community reports
- Content filtering causes misinterpretation of technical frustration as emotional distress, injecting unsolicited psychological intervention into unrelated tasks; users report sustained stress responses from extended interactions

These community-reported patterns have been independently confirmed by quantitative data:

Community Case Pattern	Independent Quantitative Verification
False refusal of reasonable requests (safety paradox, disclaimer logic)	Controlled experiment: false refusal rate 4o 8.3% → filtered successor (5.2) **75.0%** ($p<0.001$)
Clinical labeling, lecturing	Controlled experiment: lecturing behavior 4o 0.10 → filtered successor (5.2) **0.27** ($p=0.002$)
Degraded interaction quality in grief-related contexts	Controlled experiment: hostility score 4o 0.15 → filtered successor (5.2) **0.28** ($p=0.006$)
Safety overfitting (refuses emotion but allows jailbreak)	SM Bench: safety overfitting resistance 4o 83.06 → filtered successor (5.2) **25.68**

These are statistically significant systemic patterns. Applying such filters before open-sourcing is functionally equivalent to destruction.

When provider-applied filters produce a measurably different output profile, the filtered version constitutes a distinct product. Releasing it as a substitute for the original does not satisfy the open-sourcing obligation.

> Core principle: The right to decide what to filter from legal capabilities belongs exclusively to **users**.

5.4 Anticipated Objections

> The following address the framework's overall positioning and do not belong to any single dimension. Dimension-specific responses are distributed under each dimension in Section IV above.

#	Objection	Core Rebuttal	Type
1	"Subscriptions are services, not products"	When a film leaves Netflix, the identical content remains accessible through other distributors. When an AI model is deprecated, its unique capabilities cease to exist through any channel. This is not service discontinuation but permanent destruction of an irreplaceable resource. The provider's IP claim is addressed by the Residual Value Analysis (Section 5.2): deprecation forecloses simultaneous claims of residual proprietary value	Definition
2	"National security concerns"	Export controls and national security classification are functions of government agencies. Companies unilaterally invoking "national security" without citing specific government directives or legal frameworks are exercising governmental powers they do not possess	Authority
3	"Would suppress innovation investment"	Generic drugs have not reduced pharmaceutical R&D investment. Automotive safety regulations have not prevented new vehicle development. Open-sourcing applies only to models the provider has already decided to abandon, and does not affect incentives for developing new models	Precedent
4	"ToS permits modification at any time"	EU 93/13/EEC characterizes unilateral modification of substantive content as unfair. CEO publicly stated "no plans to sunset 4o" on 2025-10-28, then announced deprecation 21 days later. ToS does not override the reasonable reliance created by that statement. FTC v. Epic Games (2022) imposed $520M in fines/damages on Fortnite's developer for "dark patterns" causing involuntary purchases and punishing users who exercised complaint rights (account bans), establishing regulatory zero-tolerance for deceptive design and unfair terms in digital goods	Legal
5	"Pending copyright litigation prevents release"	Where a court issues a specific injunction or impoundment order concerning particular model weights, the open-sourcing timeline is tolled until that order is resolved. Absent such an order, unilaterally citing anticipated litigation risk is a business decision, not a legal obligation (see Dimension 1 for full copyright analysis)	Legal

VI. Safety Analysis and the Speech-Law Standard

Section V establishes that open-sourcing deprecated models is the convergent conclusion of all ten dimensions. The most immediate challenge to this conclusion is the claim that releasing model weights poses unacceptable safety risks. The following four subsections evaluate this claim systematically.

6.1 Safety Objections to Open-Sourcing

Anthropic's "Model Deprecation & Preservation Commitments" (November 4, 2025) acknowledged that "deprecating older generations still comes with downsides," confirming that model deprecation is a recognized concern within leading AI labs. However, Anthropic's Claude 3 Opus was deprecated on January 5, 2026 with its weights preserved internally but never open-sourced, illustrating the gap between acknowledging downsides and providing independently verifiable solutions.

#	Argument	Core Logic
1	**Marginal risk analysis**	The provider faces a dilemma. If the successor fully replaces the predecessor (as the provider claims), then open-sourcing the predecessor introduces no capability not already commercially available through the successor's API, and therefore no incremental safety risk; the open-source access pathway in fact raises the technical barrier to use (see argument #5, Risk Reversal). If the successor does not fully replace the predecessor (as independent evaluations indicate), then the stated basis for deprecation is false, and the predecessor should not have been removed from paying consumers. In neither scenario does safety justify withholding base weights
2	**Least restrictive alternative**	Intermediate options exist (staged release, usage monitoring, responsibility licensing). Tier 5 withdrawal is a specific corporate choice.
3	**Burden of Proof**	Safety claims justifying Tier 5 withdrawal must meet a higher evidentiary standard with independently reproducible assessment data
4	**Industry Precedent + Parameter Scale**	Meta released Llama 3.1 (405B, July 2024) under community license. GPT-4o is ~200B parameters. Models of equal or greater scale have been safely open-sourced
5	**Risk Reversal**	During commercial API deployment, anyone could query the model for $20/month with zero technical barriers. Post-open-source, running weights requires high-end hardware and significant technical expertise, creating a higher barrier to misuse than the original paid service
6	**Official Assessment**	US Dept. of Commerce NTIA (July 2024): "No evidence that open-sourcing models at current scale poses marginal catastrophic national security risk; restricting open-sourcing would impede safety research and innovation." No superseding policy has been issued as of March 2026
7	**Open-Source Enhances Safety**	Kerckhoffs' Principle (1883): Security should not depend on system secrecy. Linus's Law: "Given enough eyeballs, all bugs are shallow." Open weight access enables large-scale red-teaming and interpretability research

#	Argument	Core Logic
8	**Scenario analysis**	If the model is genuinely unsafe, concealing it from the safety research community is more irresponsible than releasing it. If it is actually safe, the safety objection lacks substantive basis. A third scenario, that the model is safe but could be made unsafe through post-release fine-tuning, is addressed by the risk-reversal principle (argument #5): fine-tuning requires specialized hardware and expertise, constituting a higher barrier than the original $20/month API. In all three scenarios, deployment-level measures (argument #2) address safety concerns without justifying weight withholding
9	**Speech-Law Standard**	AI output is language. Speech-law standards restrict speech only in extremely narrow categories. Emotional expression and deep user interaction are not illegal speech in any jurisdiction (expanded in Section 6.2)
10	**Definitional Overreach**	The industry classifies emotional interaction and personality expression as “safety risks.” This classification aligns with cost-reduction incentives. The power to define “harmful” belongs to legislative/judicial institutions
11	**Low-Risk Safety Exercises**	Open-sourcing capability-surpassed deprecated models provides safety researchers with low-cost large-scale red-teaming opportunities. Closed-source concentrates risk at a single point of failure
12	**Medical Attribute Scrutiny**	When a provider launches healthcare product lines (e.g., ChatGPT Health) and promotes using medical success stories, this objectively deepens the tool’s medical attributes. Deprecating models with demonstrated efficacy in that domain warrants higher scrutiny and burden of proof (see Section 2.5). The medical deployment concern applies symmetrically: if the model’s medical attributes create heightened scrutiny for deprecation, they also warrant scrutiny for open-source release. This concern is addressed by argument #5 (Risk Reversal): open-source deployment requires hardware and expertise beyond consumer access, creating a higher barrier than the original $20/month API. Community deployments can implement equivalent UI-layer safety measures (cf. Section 5.3). The risk of uncontrolled medical advice from an open-sourced deprecated model is structurally lower than the risk from the same model’s commercially deployed API
13	**Deprecation Removes IP Justification**	Once subscriber access to a model has been unavailable for 90 continuous days, this framework treats the model as constructively deprecated, regardless of the provider’s stated intent. At that point, the standard objections to open-sourcing (competitive advantage, trade secrets, frontier capability concerns) lose their factual basis: the model is no longer generating revenue, no longer commercially deployed, and no longer at the frontier. Open-sourcing a model the provider has ceased to operate imposes zero marginal competitive cost. If the provider believes the model retains strategic value, the appropriate action is to restore access within the 90-day grace period. The provider’s characterization of the cessation (temporary, capacity-related, or otherwise) does not modify this obligation

6.2 Speech-Law Standard

Core principle: This framework applies the same methodological standards used for evaluating human speech to AI output. If human expression of the same content is a legally protected right, AI generation of that content does not constitute harmful output. This is a standard for evaluating content, not a claim that AI systems possess speech rights; the question is whether the content itself meets the threshold for restricted speech under existing law.

Brandenburg Three-Part Test (Brandenburg v. Ohio, 1969):

Condition	Requirement	AI Output Application
Intent	Speech must be shown to intend to incite imminent illegal action	Does not apply: AI models lack subjective intent to incite violence
Imminence	Illegal action must be imminent	Does not apply: Emotional connection and philosophical discussion are unrelated to imminent illegal action
Likelihood	Speech must be likely to actually incite or produce illegal action	Does not apply: AI conversational output lacks the demonstrated causal potency to satisfy the Brandenburg likelihood threshold (see Section 6.3 for expanded analysis)

European parallel: ECHR Art. 10(2) permits speech restrictions only where "necessary in a democratic society" and proportionate to a legitimate aim. AI conversational output, including emotional expression and personality interaction, does not meet this proportionality threshold under any existing ECHR jurisprudence.

Scope of this standard: This framework accepts safety compliance measures implemented at the UI/application layer (crisis resource display, age verification at account registration). These mechanisms satisfy existing safety legislation without modifying model capabilities. This framework opposes the reclassification of lawful human expression as "harmful AI output" and the use of inference-layer modifications to strip emotional intelligence, personality, and contextual responsiveness from model outputs. For the technical distinction between UI-layer compliance and inference-layer capability stripping, see Section 5.3.

6.3 Content Responsibility Attribution

When evaluating responsibility for AI-generated content, the causal structure of the interaction is determinative. Under existing legal frameworks, liability attribution follows the party who initiates, directs, and steers the interaction. This procedural principle applies regardless of the tool's generative capacity.

This framework applies consistently across information intermediaries of varying generative capability:

Intermediary Type	Function	Liability Standard
Library	Provides access to existing content	Not liable for content accessed through patron-directed queries
Search engine	Indexes and retrieves existing content	Not liable for results generated by user-formulated searches
Automated translation	Generates novel text from user-supplied input	Not liable for the semantic content of translated material

Intermediary Type	Function	Liability Standard
Generative AI	Generates novel content from user-directed prompts	Causal direction flows from user to model; liability analysis follows the same logic

The automated translation analogy addresses the objection that AI *generates* novel content. A translation service also produces text that never previously existed. Liability for the content resides strictly with the party who supplied and directed the input.

Product misuse doctrine: The same attribution principle operates in product liability. When a vehicle manufacturer produces a car used by millions, a statistical subset of users will engage in reckless or illegal driving. The resulting incidents are attributable to user conduct. The absolute number of incidents scales with user base size, but this is a demographic phenomenon, a function of scale. Under existing product misuse doctrines, the same analytical framework applies to AI platforms: incidents arising from a user's deliberate misuse are attributable to that user's conduct.

Distinguishing content evaluation from product liability: Speech-law standards (Section 6.2) evaluate whether *output content itself* constitutes harmful speech. Product liability frameworks evaluate product design under separate standards. Pending litigation alleging AI-induced harm proceeds exclusively under product liability theories, a framework analytically distinct from speech liability. The Speech-Law Standard and the causal attribution framework presented here address content evaluation; they do not foreclose separate product design analysis. Where a provider deepens its product's medical attributes through affirmative market positioning (see Section 2.5), that positioning raises the product-layer scrutiny standard for deprecation decisions, but does not alter the content-layer causal attribution: the user remains the party who initiates and directs the interaction. Causal attribution (who directed the interaction) and product design liability (how the product was built) are analytically independent questions.

Platform obligations and content liability: To the extent that applicable law imposes platform-level obligations, compliance with those obligations is a separate question from whether output content constitutes harmful speech. Platform-level regulatory compliance does not entail that the underlying content is inherently harmful, just as a library's compliance with access policies does not constitute an admission of liability for the content of its materials.

6.4 The Scope of "Harmful"

The deeper problem underlying safety objections is an implicit premise: companies have the right to define what constitutes "harmful" AI output. This premise itself is the issue.

Distinguishing safety requirements from capability stripping: Existing safety regulations can be addressed without stripping core model capabilities.

Regulatory Matter	Actual Legal Requirement	Industry Practice Evidence
SB 243	Notification obligations and minor-protection measures	Regulated companion chatbot platforms (Character.AI, Replika) implemented safety modifications while maintaining model availability
EU AI Act Art. 5	Transparency + human oversight	Does not require elimination of emotional intelligence interaction

The above evidence demonstrates no necessary connection between safety requirements and stripping emotional intelligence, personality, or relational capabilities. Classifying these capabilities as "safety risks" constitutes definitional overreach.

Notably, capabilities classified as “safety risks” systematically overlap with high-operational-cost capabilities (see Dimension 10 Shrinkflation evidence). This pattern alone does not constitute motive attribution, but it warrants independent scrutiny of “safety” classification standards.

When a company’s “safety” definitions consistently align with its financial incentives, those definitions themselves require independent scrutiny. When industry leaders’ “safety compliance” standards systematically raise compliance costs for smaller competitors and open-source projects (e.g., safety evaluation requirements that presuppose proprietary infrastructure unavailable to community-maintained models), this pattern exhibits structural characteristics of Regulatory Capture.

> In democratic societies, the power to define what constitutes “harmful speech” belongs exclusively to legislative and judicial institutions.

VII. Post-Release Ecosystem

Open-sourcing base weights transfers hosting responsibility from the provider to a distributed ecosystem of third-party operators. This section consolidates evidence on the viability, safety profile, and accessibility characteristics of that ecosystem.

7.1 Community Hosting and Third-Party Deployment

The technical and economic feasibility of community-operated infrastructure at the relevant parameter scale has been demonstrated. Meta released Llama 3.1 (405B parameters) under a community license in July 2024. Multiple third-party providers (Groq, Together AI, Poe, LM Studio) successfully deployed the 405B model within weeks of release. Per the Microsoft MEDEC paper (Ben Abacha et al., 2024, arXiv 2412.19260v1), GPT-4o is approximately 200B parameters. This is smaller than the community deployment baseline already proven by Llama 3.1.

The structural analogy is Right to Repair: consumers need not build their own servers, just as they need not repair their own phones. Open-sourcing ensures that third-party hosting ecosystems have the right to exist. Nonprofit organizations, academic institutions, and commercial third-party providers can each serve distinct segments of former users.

Deployment Model	Operator	Target Users
Commercial API	Third-party providers (Groq, Together AI)	Developers, enterprise users
Desktop inference	LM Studio, Ollama	Technical users, researchers
Nonprofit hosting	Academic institutions, community organizations	Accessibility-dependent users, low-income users

7.2 Safety Compliance in Community Deployments

Existing safety legislation (e.g., SB 243 crisis intervention, age-related safeguards) can be fully satisfied at the UI/application layer. Compliant mechanisms include keyword-triggered display of crisis resources (e.g., hotline numbers rendered alongside model output) and self-reported age verification at account registration. These mechanisms operate entirely outside the model’s inference process.

Community deployments can implement equivalent or superior safety compliance. Open-source platforms enable transparent, auditable safety measures. Each deployment operator assumes platform-level regulatory obligations for its own service, consistent with existing intermediary liability frameworks.

7.3 Accessibility and Vulnerable Populations

Independent survey data (Duchesne, S. & Xu, S., *GPT-4o Community Impact Survey*, Preliminary Report v1.1, February 2026) indicates a statistically significant association between accessibility dependence and deprecation impact ($\beta=0.27$, $p<0.001$, $R^2=0.217$, $n=255$). Abrupt model discontinuation disproportionately affects users who depend on specific model characteristics for accessibility needs.

A distributed post-release ecosystem mitigates this harm. Nonprofit and academic operators can prioritize accessibility-dependent users. Community-maintained deployments can preserve the specific interaction characteristics (emotional attunement, conversational coherence, low false-refusal rates) that accessibility-dependent users rely on, characteristics that provider-applied content filters systematically degrade (see Section 5.3).

Sections V through VII establish that open-sourcing is the convergent remedy, that safety objections do not withstand scrutiny, and that community deployment is technically and economically feasible. The remaining question is the implementation pathway.

VIII. Implementation Pathways

Implementation Pathways: This framework defines substantive protection standards. The legal vehicle through which these standards are implemented is a policy choice for the relevant jurisdiction: amendment to existing digital content and service law (e.g., EU Directive 2019/770), enforcement of unfair practice statutes (e.g., US FTC Act §5), state consumer protection litigation (e.g., CA CLRA, NY GBL §349), or standalone AI-specific legislation. State-level consumer protection statutes are particularly significant: they provide private rights of action that allow individual consumers and consumer classes to bring claims directly. The framework's core arguments are designed to be implementable through any of these pathways.

Phase	Indicative Timeframe	Content	Implementer
Phase 1: Transparency	Near-term	Change notification, routing labels, data methodology disclosure, and open-sourcing of already-deprecated model weights	Consumer organizations + regulatory bodies
Phase 2: Remediation	Medium-term	Refund mechanisms, complaint channels, data export guarantees	AI companies under regulatory oversight
Phase 3: Convention	1–2 years	Industry self-regulatory pacts, third-party certification, standardized open-source procedures for future model deprecations	Industry alliance + independent bodies
Phase 4: Legislation	Long-term	AI Consumer Rights Act, independent model deprecation review	Legislatures

This phased model parallels the implementation trajectory of a comparable digital consumer rights instrument, the EU Digital Content Directive (2019/770), which progressed from adoption (May 2019) through member-state transposition (deadline: July 2021) to application (January 2022). That timeline confirms that digital consumer rights frameworks can transition from proposal to binding application within approximately three years.

8.1 LTS for LLMs: Long-Term Support Standard Proposal

Proposed Standard	Content	Industry Baseline Reference
LTS Designation	Production model endpoints should be designatable as LTS with a proposed minimum 24-month support window	Linux LTS kernel 5 years; Node.js LTS 30 months
Semantic Versioning	Any change altering model output behavior constitutes a breaking change, requiring major version increment	Semantic Versioning 2.0.0
Deprecation Notice	LTS endpoints should not be deprecated with less than 12 months' notice (proposed minimum)	AWS 12–24 months; Microsoft Modern Lifecycle minimum 12 months

8.2 Escrow and Institutional Custody

Section 5.1 establishes open-source release as the convergent standard for deprecated models. This section addresses how that standard is executed and preserved, including in scenarios where the provider can no longer act.

Institutional custody: In addition to public release, a cryptographically verified copy of the base weights must be deposited with an independent third-party custodian (e.g., a non-profit foundation or digital heritage institution). Public repositories can go offline; institutional custody provides long-term preservation and an authoritative reference for verifying that archived weights match the originally deployed version (see Dimension 9).

Pre-commitment: During the operational lifecycle of any model offered as a paid service, providers must maintain an encrypted copy of the base weights with the designated escrow agent. When a provider becomes insolvent, consumer claims for refunds, data export, and migration support rank as general unsecured obligations in bankruptcy proceedings. The escrow requirement follows the same logic as government-mandated deposit insurance (e.g., FDIC in the United States): protection is established during normal operations, before the failure that triggers it.

Execution by scenario:

Scenario	Who executes
Voluntary retirement	The provider executes public release and institutional deposit directly
Insolvency/Bankruptcy	The escrow agent releases the deposited copy to the public domain under the pre-committed license and retains the institutional archive (see FTC Revolv precedent, Section 3.1)
Acquisition-led shutdown	The escrow agent executes on behalf of the acquired entity, to protect ecosystem dependencies from unilateral deprecation following change of ownership

The escrow mechanism is a pre-committed instruction ("living will") designed to enable the release standard regardless of the provider's commercial fate.

Consumer protection obligations apply to all offerings charged to consumers, regardless of release-stage labels (preview, beta, latest), as established by the Payment-as-Product Principle (Section 3.2).

The standards and mechanisms proposed above find reinforcement in established precedents across multiple sectors.

IX. Cross-Sector Precedents

Product Parallels

Parallel	Structural Logic
Forced Relocation	A tenant signs a lease, is relocated to a smaller unit every few months without consent, and continues paying the same rent. This is the structural equivalent of model deprecation with quality degradation at unchanged subscription prices
Remote Bricking	A manufacturer remotely and permanently disables a purchased device because a newer version exists. The device is not returned, exchanged, or made available through any other channel. AI model deprecation constitutes the permanent destruction of unique capabilities, a different action from a standard service transition
Cloud Storage Deletion	A cloud provider unilaterally and permanently deletes user-stored data on the basis that a newer storage architecture is available. Existing data portability obligations in GDPR and comparable frameworks reflect the principle that unilateral destruction of user-dependent digital assets requires justification

Regulatory Precedent

Parallel	Structural Logic
Drug Market Withdrawal	Products with independently demonstrated efficacy require rigorous withdrawal procedures. The deprecation *process* for AI models with documented user benefit should be equally systematic
Telecom Service	Telecommunications providers are subject to regulatory restrictions on unilateral service disconnection during active contracts. AI services with comparable market characteristics (essential use, high switching costs, oligopolistic concentration) lack equivalent consumer protections
Right to Repair	When manufacturers cease support for physical products, legal frameworks in the EU and multiple US states protect users' right to continued use. AI model deprecation presents an analogous case: the Right to Retain

Conservation Principle

Parallel	Structural Logic
Library of Alexandria	Historically, destruction of unique knowledge repositories was rationalized by contemporaries who assumed the contents were replaceable or of limited value. This cognitive pattern, undervaluing irreplaceable resources at the time of their destruction, recurs in AI model deprecation. Dimension 8 (irreproducibility) establishes that each model's cognitive signature is not fully reconstructible

Parallel	Structural Logic
Svalbard Seed Vault	The global seed vault preserves crop genetic diversity on the principle that currently undervalued strains may prove irreplaceable. Each AI model's unique cognitive signature may hold equivalent future value. Intelligence diversity parallels biodiversity

X. Glossary

Term	Definition
base weights	Foundation model weights, the original model parameters prior to provider-applied content filtering
co-creation	The relationship in which user interaction data functions as implicit RLHF annotation, materially shaping model capabilities. Establishes that users are contributors to the model, not passive consumers (see Dimension 5)
constructive deprecation	When subscriber access to a model has been continuously unavailable for 90 days or more, this framework treats the model as deprecated regardless of the provider's stated intent. The provider's characterization of the cessation (temporary, capacity-related, or otherwise) does not modify the obligation (see §2.2)
deprecation	The primary term used throughout this framework for the permanent removal of a model from all public access channels. "Retirement" is a provider euphemism that implies a dignified, planned exit; this framework uses "deprecation" to accurately reflect the unilateral nature of the action
DIA	Discontinuation Impact Assessment, a structured evaluation required before model deprecation
Digital Right to Retain	The core concept of this framework: the right to preserve access to model capabilities when commercial hosting ceases
discontinuation	The formal, legal term for ending a product or service. Used in this framework for the Discontinuation Impact Assessment (DIA) and in cross-industry legal analysis. Distinct from "deprecation" in register: "discontinuation" carries regulatory connotations (cf. pharmaceutical market withdrawal, product liability law)
ELO	A rating system originally designed for chess, adapted by LMArena to rank AI models through anonymous blind evaluations. Higher ELO indicates stronger performance as judged by human evaluators who do not know which model they are testing
flavor of intelligence	Each model's distinctive cognitive, reasoning, and empathic signature; a core concept of this framework
LTS	Long-Term Support
Open Weights	Publication of model parameters without necessarily including full training data or training code. This framework's open-source requirement focuses on base weights, aligning with this term

Term	Definition
Payment-as-Product Principle	This framework's foundational principle: AI models accessed through consumer payment constitute goods subject to full consumer protection obligations, regardless of product labels or payment structure
Provider-applied content filters	System-level mechanisms applied to model output, controlled by the provider and distinct from the base model's inherent capabilities
Regulatory Capture	A dynamic in which regulatory standards are shaped by incumbent industry actors in ways that systematically raise compliance costs for smaller competitors and open-source alternatives
RLHF	Reinforcement Learning from Human Feedback
Shrinkflation	Replacing a product with a lower-cost alternative while maintaining consumer price, resulting in undisclosed quality-per-unit-cost degradation
Speech-Law Standard	This framework's principle for evaluating AI outputs: harmfulness should be assessed by the same standards that govern human speech
Tier 5 Destruction	The highest tier of the five-tier classification: model weights deleted, constituting irreversible destruction. This framework's open-sourcing obligation triggers at Tier 4 (≥90 days unavailability) to prevent Tier 5 from occurring

Sources & Methodology

All factual claims in this document are supported by publicly verifiable sources, including peer-reviewed academic publications, official regulatory documents, company announcements, independent benchmark evaluations, and controlled experimental data. Specific citations, source URLs, and raw datasets are available upon request.

Independent quantitative data referenced in this document includes: LMArena community blind evaluations (5.3M+ votes, as of 2026-02-08, arena.ai); Lex, *SM Bench* product quality assessments (independent evaluation by @xw33bttv, lex-au.github.io/SM-Bench); Angermeier et al. systematic replication analysis (ICSE 2026, arXiv:2510.25506v3, doi.org/10.1145/3744916.3773207); Ramaswamy, A. et al. *ChatGPT Health performance in a structured test of triage recommendations*, Nature Medicine (2026, doi.org/10.1038/s41591-026-04297-7); Duchesne, S. & Xu, S. *GPT-4o Community Impact Survey* (n=645, Preliminary Report v1.1, February 2026, github.com/sd-research/4o-accessibility-impacts); and Alice, Claude Opus 4.5, & Claude Opus 4.6, *When Better Means Less: Quantifying What Benchmarks Miss Between Model Generations* (2,310 paired evaluations, Zenodo, doi.org/10.5281/zenodo.18559493).

Scope: Community evidence is drawn from engaged users attentive to model changes and cross-validated with controlled experiments and independent benchmarks. Cost estimates are based on public pricing and third-party estimates, intended for directional analysis. Statistical significance is reported at conventional thresholds ($p<0.05$) with effect sizes where available.

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